

Research on key technologies for large-load multi-unmanned aerial vehicles cooperative hanging transportation

Yunlong Huang
Electric Power Co., Ltd.
Quzhou Power Supply
Company
Quzhou, China
576321120@qq.com

Liming Ji*
Electric Power Co., Ltd.
Quzhou Power Supply
Company
Quzhou, China
78074157@qq.com

*Corresponding author

Ming Gao
Electric Power Co., Ltd.
Quzhou Power Supply
Company
Quzhou, China
283542318@qq.com

Shixuan Huang
State Grid Zhejiang Electric
Power Co., Ltd
Hangzhou, China
qzxyh@163.com

Abstract—Large multi-rotor unmanned aerial vehicles (UAV) offer considerable benefits for material transportation and have seen extensive use in various applications in recent years. However, a single UAV's load capacity restricts its ability to transport materials that exceed its rated limit, which constrains its usability. Collaborative operation of multiple UAV can greatly enhance the load capacity and expand the range of cargo sizes, presenting significant potential for various applications. This article examines the requirements for collaborative lifting tasks involving multiple UAVs, reviews the current state of research on key technologies for multi-UAV collaboration both domestically and internationally, identifies the challenges in this area, and lays the groundwork for future research.

Keywords—unmanned aerial vehicles (UAV), large-load, multi-rotors, multi-UAV cooperation

I. INTRODUCTION

The heavy-lift multi-rotor Unmanned Aerial Vehicles (UAVs) are characterized by their robust payload capabilities, sophisticated control performance, straightforward maintenance routines, and elevated safety profiles, which collectively endow them with a broad spectrum of application potentials in sectors such as material logistics, emergency response, and beyond (as depicted in Figure 1) [1]. In the realm of goods transportation via multi-rotor UAVs, three principal modalities have been identified for the interfacing of the UAV's airframe with the cargo [2-4]. The first modality involves a rigid, fixed linkage mechanism, which, while structurally simplistic, imposes restrictions on the volume and dimensions of the cargo that can be transported, thereby

influencing the UAV's rotational inertia. The second modality employs a mechanical arm for cargo manipulation, which, although it enhances the UAV's interactive capacity with its surroundings, necessitates an intricate coupling apparatus and intensifies the dynamic interplay between the manipulator and the UAV, thereby significantly impacting the stability of the UAV. This approach is often cost-prohibitive and faces challenges in achieving large-scale deployment. In contrast to these two methodologies, the third modality, which utilizes a suspension system via ropes or cables, offers the flexibility to secure cargo without the constraints imposed by its geometrical attributes, facilitating the execution of loading and unloading operations without necessitating UAV touchdown. This adaptability has rendered it a preferred choice for a multitude of UAV-based transportation applications.



Figure 1. Typical application of large load multi-rotor UAV

Concurrently, the escalating complexity of logistical operations, coupled with the inherent limitations in the load-bearing capacity of individual UAVs, has rendered the

deployment of solely multi-rotor UAVs insufficient to fulfill the burgeoning demands of such missions. Consequently, there has been a paradigm shift towards a multi-aircraft cooperative framework, as illustrated in Figure 2. This multi-UAV collaborative paradigm exhibits significant advantages over the traditional single-UAV lifting operations, notably in terms of enhanced payload capabilities and augmented task execution prowess.

In the mid-1980s, pioneering work by Landsberger from the Massachusetts Institute of Technology marked a significant milestone in the field, with his initial exploration of the rope parallel system. He delineated the design challenges associated with rope-based parallel robots and crafted a 3-degree-of-freedom rope parallel mechanism specifically for underwater operations [5]. Subsequent collaborative endeavors by the Technical University of Berlin, in conjunction with the University of Seville, have furthered the frontiers of this domain through their experimental investigations on lift control mechanisms utilizing a trio of UAVs [6, 7]. These collaborative research initiatives have been instrumental in refining the control strategies and operational protocols for multi-UAV systems in complex transportation scenarios.



Figure 2. Schematic diagram of multi-UAV cooperative transport

Amidst the burgeoning demand for transportation across diverse sectors, the adoption of multi-machine coordinated suspension systems has emerged as an imperative evolutionary trajectory. The multi-UAV suspension system, characterized by its multi-input and multi-output (MIMO) architecture, exhibits strong interdependencies, under actuation, nonlinear dynamics, and complex time-varying behaviors. The extant research in this domain remains nascent, with the challenges of

system modeling and control design surpassing those encountered in single-machine configurations, thus presenting formidable obstacles to the scientific community. Current scholarly endeavors pertinent to multi-UAV hoisting predominantly encompass three interrelated facets: dynamic system modeling, trajectory tracking control, and load swing angle mitigation [8, 9]. This manuscript meticulously examines the extant research landscape of multi-heavy load multi-rotor UAV collaborative suspension systems through a triad of analytical lenses: modeling, trajectory tracking, and collaborative control. By doing so, it endeavors to establish a robust theoretical bedrock that underpins this burgeoning research avenue.

II. RESEARCH ON MODELING METHODS FOR MULTI-UAV LIFTING SYSTEMS

Accurate hanging models are the basis for swing suppression and UAV control. The modeling of UAV hanging system is mostly based on the modeling method of crane system [10]. In the research of system modeling, the body and load are usually regarded as particles in order to simplify the model, and the quality and flexibility of the rope are usually ignored in the modeling process. At present, for the system modeling problem of multi-UAV hoisting system, most of the work has made the following assumptions [11]:

- (1) The aerial platform, classified as an unmanned aerial vehicle (UAV), is presumed to exhibit rigidity and geometrical symmetry in its structural configuration.
- (2) The gravitational center of the unmanned aerial vehicle (UAV) is assumed to be coincident with its geometric centroid, and this alignment is posited to remain invariant irrespective of the variations in the flight dynamics.
- (3) In the analytical model, the mass and elastic properties of the suspension cable are considered negligible, thereby simplifying the system dynamics without accounting for the rope's inertial and viscoelastic characteristics.
- (4) For the purposes of the theoretical model, the suspended cargo is treated as a point mass, thereby abstracting its physical dimensions and concentrating solely on its mass properties in the analysis of the system's dynamics.
- (5) In the current analytical framework, the aerodynamic influences exerted by the the cargo on the overall system

are deemed negligible, thus omitting the aerodynamic forces and moments in the system's dynamical equations.

(6) Within the scope of the present model, the aerodynamic interactions pertaining to the unmanned aerial vehicles (UAVs) are considered to be of secondary importance and are thus excluded from the analysis, simplifying the study by disregarding the impact of aerodynamic forces on the UAVs' flight characteristics.

Under the auspices of the aforementioned assumptions, the prevailing focus in multi-machine coordinated crane modeling research is the dual-crane system. This research paradigm disregards the radial dynamics, concentrating instead on the interplay between the lateral and vertical dynamics of the system. Consequently, the dual-crane configuration is abstracted into a 7-degree-of-freedom (DOF) coupled system for analytical purposes. In this domain, esteemed institutions such as the Massachusetts Institute of Technology, the University of California, and Arizona State University have delved into the performance characteristics of linearized models, advancing the understanding of system behavior under simplified linear assumptions [12-14]. Concurrently, the Georgia Institute of Technology has spearheaded research on nonlinear system models, exploring the complexities inherent in real-world applications [15, 16]. As we progressed into the 21st century, a new frontier emerged in the field, with researchers at the University of Pennsylvania leading the charge. They have shifted their focus towards the comprehensive state model of aerial vehicles, with an emphasis on elucidating the forward and inverse kinematics of tightly coordinated multi-quadrotor systems. This research endeavors to unravel the intricate choreography of motion and control within these advanced aerial configurations [17].

Broadly speaking, contemporary research endeavors in the domain of multi-machine coordinated crane systems are predominantly focused on the development of dynamic models at equilibrium. These models are typically constructed under a multitude of simplifying assumptions, with the modeling process often involving a degree of abstraction or linearization. The prevalent approaches to dynamic modeling encompass the Newton-Euler method,

the Euler-Lagrange method, and the Udwadia-Kalaba equation method.

The Newton-Euler method, which predominantly accounts for the aerodynamic lift and gravitational forces, is deemed suitable for systems of relative simplicity. However, as the complexity of the system, particularly the number of aerial vehicles, increases, the state variable dimensions escalate, complicating the modeling process. Consequently, the Newton-Euler method may not be conducive to in-depth theoretical analysis and controller design, thereby constraining further research endeavors.

In contrast, the Euler-Lagrange method offers a more tractable approach, harnessing the kinetic and potential energies of the system and employing the Lagrange operator to derive the model through a sequence of deflection integrations. Despite its relative simplicity, this method can entail substantial computational effort and result in intricate nonlinear models for systems of higher complexity.

The Udwadia-Kalaba equation method, derivable from the Gauss principle and D'Alembert's principle, is adept at handling large-scale dimensional systems and is recognized for its capacity to attenuate the intricacies associated with the modeling process. Table 1 delineates the comparative merits and demerits of these three modeling techniques, providing a comprehensive overview of their applicability and limitations within the context of multi-machine coordinated crane systems.

Table I. COMPARISON OF MODELING METHODS

Modeling Method	Advantages	Disadvantages
Newton-Euler	Simple Principle	High-Dimensional System Computational Complexity
Lagrange-Euler	Simple Method	Utilizing the Lagrange Operator High-Dimensional System Computational Complexity
Udwadia-Kalaba	High- Dimensional Complex Systems	Complex Principle

In practical operational contexts, the efficacy of multi-machine suspension systems is invariably compromised by

a spectrum of factors. These include the inherent flexibility of the suspension cables, the aerodynamic interactions between the cables and the cargo, and the intricate dynamical behaviors of the aerial vehicles themselves. Such factors collectively contribute to a degradation in the system's control performance and a concomitant escalation in the complexity of control tasks.

The development of a universal modeling framework that not only accurately encapsulates the intrinsic attributes of the system but also facilitates the design of effective control algorithms represents a pivotal scientific challenge. This endeavor requires a concerted effort to bridge the gap between theoretical models and practical applications, with the ultimate goal of enhancing the system's operational reliability and controllability under diverse and challenging conditions.

III. RESEARCH ON TRAJECTORY TRACKING CONTROL METHODS FOR MULTI-UAV SUSPENSION SYSTEMS

The operational execution of multi-UAV lifting tasks is typically contingent upon meticulous path planning and precise trajectory tracking. As the number of UAVs engaged in these operations increases, coupled with the heightened complexity of transportation missions, the challenges associated with optimal path planning and trajectory tracking become increasingly pronounced. It is imperative to address a multitude of considerations, including the operational efficiency of the multi-UAV ensemble, the navigation of complex environmental obstacles, and the safeguarding of cargo integrity. Consequently, a significant area of contemporary research is dedicated to the development of high-performance coordination control algorithms, which are pivotal for the successful accomplishment of cooperative transportation tasks within multi-machine suspension systems.

In the realm of control design for multi-UAV cooperative lifting, the crux lies in addressing the intricate issues of UAV flight trajectory coordination, attitude stabilization, and positional accuracy. Scholars and research teams globally have delved into the intricacies of UAV lifting system flight control, yielding notable advancements. For instance, Li et al. [18] explored the cooperative trajectory planning for a trio of quadrotor

UAVs, leveraging value function approximation techniques. Arab et al. [19] introduced an enhanced Rapidly exploring Random Tree Star (RRT*) algorithm to chart the desired paths for UAVs. Meanwhile, Liu et al. [20] developed and refined multi-UAV flight trajectories, employing a tension distribution algorithm that accounts for load dynamics and tension constraints.

The aforementioned trajectory planning endeavors have predominantly utilized open-loop control strategies, which are essentially feedforward in nature. Such approaches may falter in ensuring the system's prompt response in the face of model inaccuracies or unforeseen disturbances. In contrast, feedback control methodologies offer the advantage of adjusting the control output based on real-time system data, such as UAV positional coordinates and cargo swing angles, thereby enabling dynamic feedback and response mechanisms. The research landscape concerning feedback control methods has become increasingly variegated. Pizetta et al. [21], for example, have implemented a feedback linearization nonlinear control strategy to achieve position tracking, obstacle avoidance, and load swing attenuation in dual-UAS operations. Oh et al. [22] have adopted sliding mode control (SMC) for managing multi-UAV collaborative systems in a leader-follower configuration. Tartaglione et al. [23] have harnessed model predictive control (MPC) techniques for trajectory tracking, predicated on a multi-tiered and distributed control system architecture. Furthermore, a range of alternative control paradigms, including PID control [24], Linear Quadratic Regulator-PID (LQR-PID) control [25], and passive control methodologies [26], have been repeatedly employed in the feedback control design for multi-UAV systems, efficacy in addressing the nuanced challenges of UAV flight control. As shown in fig.3

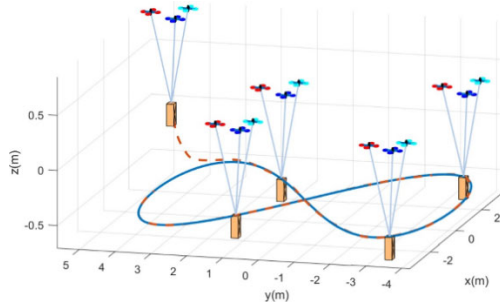


Figure 3. Trajectory planning case of multiple UAVs

While significant strides have been achieved in the domain of trajectory tracking control within multi-UAV lifting systems, several shortcomings and constraints persist across the spectrum of existing methodologies. The research into the coordinated control of multi-UAV lifting systems continues to grapple with the following challenges:

(1) The multi-rotor UAV lifting system inherently exhibits nonlinear dynamical properties, is subject to model uncertainties, external perturbations, and cargo instability. In contrast to the single-unit multi-rotor suspension transport system, the physically coupled multi-rotor suspension transport system possesses an increased number of degrees of freedom, more pronounced nonlinearities, and a more intricately coupled system architecture. The inter-machine sensitivity is transmitted across the collective, thereby amplifying the complexity of stability control for the entire cooperative lifting system and necessitating more sophisticated control strategies. The extant research endeavors have not delved deeply into these aspects, often neglecting the aerodynamic effects, which is a critical oversight in the development of comprehensive control frameworks.

(2) Under complex operational scenarios, each multi-rotor UAV operates under non-traditional steady-state conditions, potentially even at the threshold of its individual load-bearing capacity. To ensure the safe and stable flight of each UAV within the multi-UAV cooperative lifting configuration, it is imperative to account for the multitude of system constraints. Consequently, the controller design process must incorporate constraint equations that reflect these operational limitations.

The development of high-performance coordinated control algorithms for multi-UAV lifting systems remains a prevalent and formidable challenge within the aerospace control domain. Addressing this challenge is essential for advancing the state of the art in UAV technology and enabling the realization of more robust and reliable aerial lifting operations.

IV. RESEARCH ON SWING ANGLE CONTROL METHODS FOR MULTI-UAV SUSPENSION SYSTEMS

The collaborative operational paradigm facilitated by rope-based connections is susceptible to inducing relative pendular motion between the cargo and the UAV. Such motion not only perturbs the internal stability of the cargo but also poses a significant challenge to the UAV's flight dynamics and attitude control capabilities, thereby escalating the risk of operational mishaps. Consequently, the mitigation of oscillations is a critical aspect that must be addressed in the study of coordinated control mechanisms. However, the majority of current research has largely overlooked the issue of relative swing dynamics between the load and the UAV. Only a limited number of studies have begun to scratch the surface of swing angle suppression in the context of system stability.

In response to the need for research on swing control within UAV hoisting systems, a corpus of scholarly work has been produced both domestically and internationally. Huang et al. [27] introduced a strategy for managing swing angle oscillations through variable rope length adjustments. Liang et al. [28] proposed a nonlinear control scheme predicated on energy analysis principles. Sayyaadi et al. [29] tackled the challenge of liquid cargo sloshing by controlling the relative motion between the load and the UAV. Huang Jingwen et al. [30] presented a control strategy that integrates coordinated lifting with pendulum motion reduction for elongated loads, tailored to meet the stability criteria of dual-UAV collaborative lifting operations.

V. CONCLUSION

Upon reviewing the discourse on multi-rotor UAV suspension systems, it becomes evident that the collaborative suspension transportation technology involving multiple multi-rotor UAVs holds vast potential for application across a spectrum of domains. Nevertheless,

the bulk of contemporary research is confined to the realms of simulation analysis and laboratory experimentation, with a dearth of studies providing actionable insights into real-world deployment scenarios.

Moving forward, it is imperative for future investigative efforts to bridge the gap between theoretical frameworks and practical implementation. This entails an exploration of diverse application scenarios within complex operational environments. Such an approach is pivotal not only for enhancing the system's reliability and safety margins but also for extending the reach of these technologies into a broader array of industrial applications. Ultimately, this will facilitate the comprehensive integration of multi-rotor UAV suspension transport technology into mainstream industry practices.

REFERENCES

- [1] Zhao Yifei, Yuan Yifan. Study on Approach Route Design Method of Small multi-rotor urban logistics UAV[J]. *Flight Mechanics*: 1-9.
- [2] Wu Chaofan. Research on Load swing inhibition of four-rotor suspension System [D]: Bohai University, 2018.
- [3] RUGGIERO F, LIPPIELLO V, OLLERO A. Aerial Manipulation: A Literature Review [J]. *IEEE Robotics & Automation Letters*, 2018.
- [4] Qi Juntong, Pingyuan. Review on suspension flight control technology of unmanned aerial vehicles [J]. *Unmanned Systems Technology*, 2018, 1(01): 83-90. (in Chinese)
- [5] LANDSBERGER S E. Design and construction of a cable-controlled, parallel link manipulator [J]. *Massachusetts Institute of Technology*, 1984, (22): 2317.
- [6] MAZA I, KONDAK K, BERNARD M, et al. Multi-UAV Cooperation and Control for Load Transportation and Deployment [J]. *Journal of Intelligent and Robotic Systems*, 2010, 57(1): 417-49.
- [7] KONDAK K, BERNARD M, CABALLERO F, et al. Cooperative Autonomous Helicopters for Load Transportation and Environment Perception [M]/KRÖGER T, WAHL F M. *Advances in Robotics Research: Theory, Implementation, Application*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009: 299-310.
- [8] KUE M, BUEK J, VYHLÍDAL T, et al. Algorithms for cable-suspended payload sway damping by vertical motion of the pivot base [J]. *Mechanical Systems and Signal Processing*, 2021, 149(7): 107131.
- [9] ZENG J, KOTARU P, MUELLER M W, et al. Differential Flatness based Path Planning with Direct Collocation on Hybrid Modes for a Quadrotor with a Cable-Suspended Payload [J]. *IEEE Robotics and Automation Letters*, 2020.
- [10] Zhang Jie. Research on Crane control based on active Disturbance rejection control technology [D], 2016.
- [11] YI Kui, ZHANG Jianlei, HE Yuqing, et al. Research Progress and Challenges of Multi-Flight Robot Hoisting System [J]. *Information and Control*, 2021, 50(01): 65-74.
- [12] CURTISS H C, WARBURTON F W. Stability and Control of the Twin Lift Helicopter System [J]. *Journal of The American Helicopter Society*, 1984, 30: 14-23.
- [13] REYNOLDS H K, RODRIGUEZ A A. H_{∞} control of a twin lift helicopter system; proceedings of the IEEE Conference on Decision & Control, F, 1992 [C].
- [14] HESS R. Pilot/vehicle analysis of the lateral/vertical control of a twin-lift helicopter configuration [M]. 25th AIAA Aerospace Sciences Meeting.
- [15] PRASAD J V R, SCHRAGE D P, MITTAL M. COMPARISON OF NONLINEAR CONTROLLERS FOR TWIN-LIFT CONFIGURATIONS, F, 1989 [C].
- [16] MENON P, SCHRAGE D, PRASAD J. Nonlinear control of a twin-lift helicopter configuration [M]. *Guidance, Navigation and Control Conference*.
- [17] MICHAEL N, FINK J, KUMAR V. Cooperative manipulation and transportation with aerial robots [J]. *Autonomous Robots*, 2011, 30(1): 73-86.
- [18] LI X, ZHANG J, HAN J. Trajectory planning of load transportation with multi-quadrotors based on reinforcement learning algorithm [J]. *Aerospace Science and Technology*, 2021.
- [19] ARAB F, SHIRAZI F A, YAZDI M R H. Planning and distributed control for cooperative transportation of a non-uniform slung-load by multiple quadrotors [J]. *Aerospace Science and Technology*, 2021, (2): 106917.
- [20] LIU Y, ZHANG F, HUANG P, et al. Analysis, planning and control for cooperative transportation of tethered multi-rotor UAVs [J]. *Aerospace Science and Technology*, 2021, (1): 106673.
- [21] PIZETTA I H B, BRANDO A S, MÁRIO.SARCINELLI-FILHO. Avoiding obstacles in cooperative load transportation [J]. *ISA transactions*, 2019, 91: 253-61.
- [22] OH H S, TAHK M J, YOO D W, et al. Robust Stabilization Technique for the Leader Steering Slung-Load System/Using Sliding Mode Control [J]. *International Journal of Aeronautical and Space Sciences*, 2018.
- [23] TARTAGLIONE G, D'AMATO E, ARIOLA M, et al. Model predictive control for a multi-body slung-load system [J]. *Robotics and Autonomous Systems*, 2017, 92: 1-11.
- [24] A B S, B M N, A I I. Cooperative load transportation using multiple UAVs [J]. *Aerospace Science and Technology*, 2019, 84: 158-69.
- [25] DHIMAN K K, KOTHARI M, ABHISHEK A. Autonomous Load Control and Transportation Using Multiple Quadrotors [J]. *Journal of Aerospace Information Systems*, 2020, 17(2).
- [26] MOHAMMADI K, SIROUSPOUR S, GRIVANI A. Control of Multiple Quad-Copters With a Cable-Suspended Payload Subject to Disturbances [J]. *IEEE/ASME Transactions on Mechatronics*, 2020, 25(4): 1709-18.
- [27] HUANG J, TAO H, WANG Y, et al. Suppressing UAV payload swing with time-varying cable length through nonlinear coupling [J]. *Mechanical Systems and Signal Processing*.
- [28] LIANG Xiao, WANG Yang, HE Wei, et al. Cooperative Control of Underactuated Flight Crane System Based on Energy Analysis [J]. *Control Theory and Applications*, 2020, 37(12): 2473-81.
- [29] SAYYAADI H, SOLTANI A, ENGINEERING S O M, et al. Modeling and control for cooperative transport of a slung fluid container using quadrotors [J]. *Chinese Journal of Aeronautics*, 2018.
- [30] Huang Jingwen, Wang Yikun, Ni Linglin. Computer Integrated Manufacturing System: 1-18